

Zooming to Class? Experimental Evidence on College Students' Online Learning during COVID-19[†]

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One persistent question in higher education is the efficacy of online education. In the fall of 2020, we randomized 551 West Point students in a required introductory economics course across 12 instructors to either an online or in-person class as a response to the COVID-19 pandemic. Final grades for online students dropped by 0.215 standard deviations, a result apparent in both assignments and exams and largest for academically at-risk students. A postcourse survey finds that online students struggled to concentrate in class and felt less connected to their instructors and peers. Our results show detrimental effects for online learning. (JEL A22, I12, I23, I26)

Online learning has increased dramatically on college campuses with little evidence of whether it helps students. Over the last few decades, many technologists have predicted that online education would disrupt traditional markets and education delivery (Christensen and Eyring 2011; Crow and Dabars 2015). Online learning potentially could increase access for students who cannot come to a physical campus while decreasing costs (Deming et al. 2015; Deming, Lovenheim, and Patterson 2019; Cowen and Tabarrok 2014) and increasing a university's enrollment.

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However, estimating effects of online learning on student outcomes is difficult because of selection.

During the fall 2020 semester, the US Military Academy at West Point switched some students into online instruction to maintain social distancing and other safety measures essential to slowing the spread of COVID-19. As a result, we randomized 551 students across 12 instructors in 36 class sections in a required Principles of Economics course at West Point (Kofoed et al. 2024). This experimental setting allowed us to estimate differences in student outcomes between online and in-person instruction.

In our experiment, the syllabus, graded events, homework assignments, and exams were identical for online and in-person classes. We also offered both online and in-person class sections for each class hour the course was available, and we randomly assigned students within the hour into each modality. Both online and in-person class sections met during their assigned time slots, and instructors taught synchronously regardless of modality. The academy required attendance in both modalities (with cameras on for online classes). Each instructor taught both online and in-person sections, and the academy randomly assigned students within instructor to control for teaching quality.

We find that online education lowered a student's final grade by 0.215 standard deviations, a statistically significant result driven by the students with below-median academic ability. We also compare the effect of online instruction across types of graded events throughout the semester, academic ability, and student demographics. We find that male students and those who attended a year of community college (the US Military Academy Preparatory School, or USMAPS) and were prior enlisted did worse online. We conduct a postcourse survey and find that online students had reduced concentration and felt a deteriorated relationship with their instructor and peers.

The main difficulty of estimating the efficacy of online courses is overcoming selection bias of students self-sorting into modality of instruction. There are a few studies that use quasi- or pure experimental variation, but the results in each study are mixed, finding either that students do worse in online classes (Bettinger, Fox, et al. 2017; Joyce et al. 2015; Figlio, Rush, and Yin 2013; Alpert, Couch, and Harmon 2016; Cacault et al. 2021; Bueno 2020) or that there is no difference between teaching modalities (Coates et al. 2004; Russell 1999).

Bacher-Hicks and Goodman (2021) argue that quasi-experimental variation from the switch to online learning from the COVID-19 pandemic is confounded by factors such as parental job losses, illness, student anxiety, and social factors that would violate the exclusion restriction. Given the policy importance of this question, there are a few papers that estimate the learning loss during the pandemic using either difference in differences (Bird, Castleman, and Lohner 2022) or school district fixed effects with standardized test scores (Jack et al. 2023). While these studies find a negative effect, they may suffer from violations of the exclusion restriction since external factors such as unemployment or infection are correlated with both the switch to online learning and a reduction in grades.

Given the experimental setting, we argue that these findings are causal. However, there may be some limits to generalizability. First, our findings do not indicate that all online learning is detrimental but perhaps that online learning that simply ports

the traditional classroom directly into a virtual environment may worsen learning. Also, West Point classes are smaller with more individualized attention than large lecture halls used in previous experiments such as Figlio, Rush, and Yin (2013), so the online switch may have caused instructor-student relationships to deteriorate. West Point students had regular internet access, were not allowed to work off campus, and had free access to room and board. West Point cadets are also high achieving, with median SAT scores greater than those of public flagship universities but less than those of top ten universities. Our finding of detrimental effects that are concentrated in the academically weakest students are particularly concerning given that West Point cadets are also high achieving relative to most four-year college students.

Our paper contributes to the literature in several ways. First, our randomized controlled trial does not depend on volunteers since Principles of Economics is a required course, eliminating the concern about selection bias into the experiment. Also, West Point randomizes the semester in which a student takes any of their required classes, meaning that there is no self-selection into or out of this economics course in a particular semester. Next, we randomized students across 12 instructors, with each instructor teaching a balanced slate of online and in-person class sections. This feature allows us to control for instructor attributes like experience and comfort with either teaching modality.

I. Institutional Details

The US Military Academy, situated in West Point, New York, is a public liberal arts college and the nation's oldest military academy. West Point's primary mission is to educate future officers who go on to serve in the US Army. Students accepted at West Point undergo a highly selective process and come from all 50 states and several US territories and foreign militaries. The median SAT math score is 650, which is similar to state flagship institutions and selective liberal arts colleges such as Williams College, Virginia Tech, the University of Michigan, and Davidson College. In online Appendix Table A1, we show that West Point cadets tend to have higher SAT and ACT scores than those of students at the average public flagship but lower than those of students at top ten ranked universities.

To gain acceptance to West Point, an applicant must secure a nomination from her member of Congress or a senator, pass a physical fitness exam, and show leadership potential. There is no tuition at West Point, and students receive free housing, room, board, and a stipend for book and living expenses. The 47-month experience between reception day and graduation comprises military and leadership training, academic rigor, and physical challenges. Students graduate West Point with a bachelor of science degree but can major in a number of topics as diverse as English, management, mechanical engineering, and economics.

During the onset of the COVID-19 pandemic, West Point administrators needed to balance both academic and military training. Students at West Point were still required to complete 24 required "core courses," including economics, and attendance remained mandatory for each class regardless of modality.¹ Our experiment

¹ Even before the pandemic, West Point students (along with their peers at the US Air Force Academy and US Naval Academy) had little control over their academic schedules and peer networks. West Point also randomly

took place in a required Principles of Economics class.² This course is similar to introductory courses taught at other universities, and the vast majority of enrollees were second-year students. While some conditions may differ from other colleges, responses to COVID-19 by West Point leadership (i.e., 50 percent classroom capacities, pivoting courses online, testing and social distancing measures) are similar to what other college administrators faced.

II. Data and Summary Statistics

The data from this experiment come from three sources. The first source is class grades throughout the fall 2020 semester. West Point requires that each course have a unified curriculum with the same graded events across instructors and class sections, and instructors report all grades for each assignment into a central system. We used these grades to show how online education affected multiple types of academic performance, including daily homework, problem sets, midterms,³ and a final exam.

We also received data from the Office of the Dean⁴ regarding student demographics and pre–West Point academic achievement. We combined these data to ensure the internal validity of our experiment. Finally, we administered a postcourse survey to the students. This survey was voluntary, and students received extra credit for completion. Four hundred two students completed the evaluation for a response rate of 72.95 percent. There was also coverage for all class sections and instructors. Of the completed surveys, 60.45 percent were online, compared to 61.16 percent in the full sample, a result that was not statistically significant. Regarding the observable characteristics, survey respondents were slightly more likely to be female (25.37 versus 22.88 percent) and less likely to be Black (12.19 versus 14.41 percent) but were otherwise balanced against the whole sample with a joint *F*-stat of 3.76. Thus, while we do not have perfect compliance, the subsample should be representative of the whole. Researchers had sole access to the responses, and students provided their identification numbers so we could link responses to the other two datasets.

Table 1 shows summary statistics for student demographics and outcomes. Panel A displays student demographics. Our sample is representative of the West Point student body, with 23 percent of our sample female, 14 percent Black, 3.3 percent Hispanic, and 5.6 percent Asian. One of the key pillars of a student's education is physical training, and West Point recruits heavily for intercollegiate and club sports; thus, 29 percent of our sample are NCAA athletes. West Point also hosts USMAPS,

assigns students to academic instructors (Carrell, Page, and West 2010; Carrell and West 2010; Mansour et al. 2022), the semester they take a general education course (Patterson, Pope, and Feudo 2023), class hours (Carrell, Maghakian, and West 2011; Haggag et al. 2021), roommates (Jones and Kofoed 2020; Kofoed and Jones 2023), social networks called “companies” (Lyle 2007, 2009; Brady, Insler, and Rahman 2017), final exam periods, and military mentors (Kofoed and McGovney 2019).

²The Principles of Economics class consists of concepts from introductory microeconomics, macroeconomics, and personal finance, with all instructors using *Principles of Economics*, eighth edition (Mankiw 2018).

³Offering the exam online for both modalities would also ensure that if students participated in academic dishonesty (as documented by Bilen and Matros 2021), then there would not be differential opportunities for this behavior between treatment and control. In our setting, there was equal opportunity for this behavior across treatment and control. There were two versions of the exam, one for each day the exam was offered. We included class day fixed effects to account for this difference.

⁴The dean of the academic board at West Point is a one-star general and serves as the chief academic officer at the academy. This position is similar to a provost.

TABLE 1—SUMMARY STATISTICS COMPARING FULL SAMPLE AND IN-PERSON AND ONLINE CLASSROOMS

	Full sample mean/SD (1)	In-person mean/SD (2)	Online mean/SD (3)	Difference (columns 3–2) b/SE (4)	F-stat F-stat/p-value (5)
<i>Panel A. Student characteristics</i>					
Online	0.612 (0.488)	0.000 (0.000)	1.000 (0.000)		
Female	0.230 (0.422)	0.238 (0.427)	0.226 (0.419)	−0.013 [0.037]	0.06 (0.800)
Black	0.140 (0.347)	0.140 (0.348)	0.139 (0.347)	−0.001 [0.030]	0.00 (0.956)
Hispanic	0.033 (0.178)	0.042 (0.201)	0.027 (0.161)	−0.015 [0.016]	1.54 (0.215)
Asian	0.056 (0.231)	0.065 (0.248)	0.050 (0.219)	−0.015 [0.020]	0.02 (0.899)
NCAA athlete	0.290 (0.454)	0.327 (0.470)	0.267 (0.443)	−0.060 [0.040]	1.15 (0.284)
Prep school	0.142 (0.349)	0.140 (0.348)	0.142 (0.350)	0.002 [0.031]	0.04 (0.848)
Prior enlisted	0.169 (0.375)	0.182 (0.387)	0.160 (0.367)	−0.022 [0.033]	0.53 (0.466)
CEER score	623.806 (73.377)	624.944 (77.155)	623.083 (70.981)	−1.861 [6.419]	0.41 (0.522)
<i>F-stat for joint significance for all covariates</i>					0.95 (0.473)
<i>Panel B. Outcomes</i>					
Final grade (standardized)	0.000 (1.000)	0.144 (0.980)	−0.092 (1.003)	−0.236 [0.087]	
Final grade (percent)	83.367 (6.998)	84.38 (6.860)	82.73 (7.018)	−1.650 [0.608]	
Observations	551	337	214	551	

Notes: This table shows differences in standardized course grades and covariate balance across treatment and control groups. Means are in the first row with standard deviations in parentheses. Column 4 shows covariate balance across treatment and control with a difference-in-means test with standard errors below in brackets. Column 5 displays *F*-stats for a regression with the covariate of interest, including instructor, class day, and class hour fixed effects. In addition to demographic covariates, we show differences for a cadet's CEER score as a measure of academic ability. "CEER score" is an index combining measures of academic potential such as high school GPA, high school class rank, SAT score, and ACT score. In the last row of Panel A, we display the joint *F*-stat for a regression with all covariates and fixed effects. Panel B shows similar comparisons between treatment and control for outcomes. We had 12 instructors teaching a total of 38 class sections.

which offers community college–like courses for prospective students who may need some additional academic development before applying to the academy. In our sample, 14.2 percent of students were USMAPS graduates. Finally, West Point offers a program for enlisted soldiers with leadership potential to attend West Point, earn a bachelor's degree, and commission as an officer. Of the students in our sample, 16.9 percent were prior enlisted.

Finally, it is helpful to understand the academic ability of students before their attendance at West Point. To compare students from across the country, West Point derived an index called the College Entrance Examination Rank (CEER). The CEER score is a composite of the SAT, ACT, and high school GPA and has a maximum

score of 800.⁵ The average CEER score in our sample is 623. For comparison's sake, the average SAT math score of our sample is 665.81. The average grade across all sections of the course was 83.36 percentage points.

III. Experimental Setting and Random Assignment

Like many colleges and universities during the COVID-19 pandemic, West Point limited classroom capacity to incorporate social distancing measures. However, unlike other institutions, it still needed to carry out its unique mission to provide physical and military training, and thus all students returned to campus for the fall 2020 semester. Around one-half of the faculty are military faculty members who are selected by West Point from the regular army, who then attend graduate school and teach at the academy for two to three years. These officers generally teach the required introductory classes and are vital for student mentorship (Carrell, Page, and West 2010; Jones and Kofoed 2020; Mansour et al. 2022). West Point's faculty also include permanent military and civilian professors with PhDs who teach the introductory courses but focus their efforts on upper-division courses.

West Point students have little control over their daily academic schedules, and this policy did not change during the COVID-19 pandemic. We received permission to use this already existing random assignment to assign students to either an in-person or online class section. The only exception to our randomization was NCAA athletes, whom the academy prioritizes for early morning classes. However, within class hour, we randomized these students into either an in-person or online class section. Thus, our randomization is potentially valid conditional on day and class hour. Students could not appeal for a different day, time, or modality, and they could not opt to take the course another semester.

In addition, to allow for in-person instruction, each instructor agreed to teach their four-section teaching load⁶ half online and half in person. West Point classes meet roughly every other day, so instructors taught no more than two class sections per day. This environment created counterfactual classrooms where one could see a given instructor teach both in person and online.⁷ This setting allowed us to control for instructor talent, experience, quality, and familiarity with the course material. We could thus control for instructor intangibles and isolate the treatment effect of being in an online course by including instructor fixed effects. The authors chose which class sections were online, and the registrar's office randomly assigned students to class sections. Thus, instructors did not have discretion over whether a given class section would be online.

We used online software to create and execute all graded events throughout the semester. Students completed daily homework assignments that the software graded for completion as opposed to accuracy. All students took their midterms

⁵Hanser and Oguz (2015) show that the CEER score is actually highly predictive of a prospective cadet's probability of graduation and academic success but not necessarily of early promotion as an army officer.

⁶Four instructors taught only two sections, one online and one in person. Two instructors taught three sections, with two online and one in person.

⁷There were two faculty members, however, who either taught all their classes online or all in person. We dropped students enrolled in these classes from our experiment for a cleaner estimate. We also dropped students who disenrolled from West Point before the first midterm. However, our results are robust when we include these students in the sample.

and final exams online. These exams contained multiple-choice questions and software-generated graphing exercises. The software graded all assignments, and instructors had no discretion over grades, thus eliminating any potential differences in grading between modalities, such as an instructor being more lenient to their online students.⁸

Given the classroom constraints, each in-person and online class allowed for 12 and 18 students, respectively. Class size differences were a potential threat to our identification given that they could confound the treatment. Thus, of our sample of 551 students, 337 were included in an online section and 214 were in an in-person section. While class size effects are well studied in K–12 education (e.g., Angrist and Lavy 1999; Krueger and Whitmore 2001), there is less evidence of a class size effect in higher education (in particular, online higher education). Bettinger et al. (2017) use randomized class sizes in an online setting. They find little evidence of a class size effect in online classes. Essentially, for our effects to be explained by class size, they would need to be around 12 times as big as the authors' findings. We provide further evidence in online Appendix Section 4.2 by leveraging random assignment to class section and show that variation in class size does not change our effect.

Most class hours were balanced with respect to the number of online or in-person sections except for the early morning (7:55 AM) class hour, which offered more in-person than online sections. However, for every class hour the course was offered, both teaching modalities were present and the academy randomly assigned students between the formats. In total, we randomly assigned 551 students across 12 instructors for a total of 36 classrooms.

Table 1 also compares students assigned to online or in-person class sections. Column 2 shows means and standard deviations for demographic characteristics for students in online sections, column 3 shows similar summary statistics for students in in-person sections, and column 4 shows differences in means tests and whether they are statistically significant. Column 5 regresses being in an online class on a given covariate and instructor, class day, and class hour fixed effects. We report *F*-stats and their respective *p*-values from each estimated model.

Panel A shows balance on student demographics, including gender, race, and academic ability. Differences between teaching modalities are small (around 1 percentage point for female, Hispanic, and Asian students and one-tenth of 1 percent for Black students) and statistically insignificant. Regarding other student characteristics, we find that NCAA athletes are 6 percentage points more likely to be assigned an in-person section because they are assigned early morning classes to afford training in the afternoon and there were more in-person sections than online sections in the early morning hours. This difference, however, is not statistically significant. We find that former prep school students are 0.002 percentage points more likely to be in an online class, prior enlisted students are 2.2 percentage points more likely to be in an in-person class, and the average CEER score is 1.861 points (from

⁸ Altindag, Filiz, and Tekin (2021) use a student and instructor fixed effects model to study the efficacy of online education before and during the COVID-19 pandemic. They found that while in-person students outperformed online students, grade inflation among online class sections could result in the opposite result.

a mean of 622) more in online classes. Column 5 confirms these findings given that none of the F -stats are statistically significant.

Panel B shows differences in outcome variables, including standardized final grades defined as total points earned over a possible 1,000. We find that students in online class sections earned a grade 0.234 standard deviations lower than those in in-person classes (around 1.638 percentage points), or around half of a $+/ -$ grade. This result is statistically significant in a difference-in-means test. Given our randomization, this effect is potentially causal, but we test for robustness by adding a host of fixed effects and covariates.

IV. Empirical Approach

Given the random assignment of students to either a class section taught online or in person, the econometric model is straightforward. We estimate the following regression to calculate the effect of online learning on student outcomes:

$$(1) \quad y_{ijdt} = \beta_0 + \beta_1 \text{online}_{ijdt} + \gamma_j + \xi_d + \phi_t + X_{ijdt} + \epsilon_{ijdt},$$

where y_{ijdt} is the score that student i earns on a graded event (midterm, homework, final exam, final grade) from instructor j in a class taught on day d at class time t . β_1 is a parameter that estimates the causal effect of taking the class online as opposed to in person. We include γ_j , or instructor fixed effects. These instructor fixed effects will control for instructor attributes that are invariant across teaching modalities, including teaching experience, familiarity with the material, and personality. Given West Point's alternating day 1 or day 2 class schedules, we include class day fixed effects (ξ_d) to control for different exam versions and instructor "learning by doing." Next, we include time of day fixed effects (ϕ_t) to control for differences in NCAA athlete class time assignment and any student performance differentials attributed to when they take the class. Finally, we include a matrix (X_{ijdt}) of covariates such as gender, race, NCAA athlete, prior military service, and prep school graduate.

Our paper is unique in that instructors taught half their sections online and the other half in person. This assignment allowed similar students to see a given instructor in both modalities. Thus, the instructor fixed effects simulate an experiment where a hypothetically identical student would learn from an instructor in person and then compare that student to a counterfactual who only saw the instructor online. Since the curriculum is uniform and graded events are identical in content and delivery, the only difference in the experiment would be the mode of instruction. One exception is that instructors used different exam versions across class days, requiring us to adjust our estimates using day fixed effects. One limitation to this setting is that instructors could have either taught differently in an online environment or were more uncomfortable online than they were in person. Thus, the treatment effect could be capturing both changes in instruction inherent in an online environment and differences in instructor practices between the modalities. While our experiment cannot disentangle these two effects, it is representative of the experiences of many instructors across higher education during the pandemic.

For inference, we would normally adjust our standard errors for correlation between students in a given classroom by clustering at the classroom level. However, we find that the clustered standard errors are smaller than the robust standard errors, a sign of a small cluster problem. To address this concern, we follow the findings of Cameron, Gelbach, and Miller (2008) and estimate wild bootstrapped standard errors along with robust standard errors. We report p -values for inference with both the robust standard errors and wild bootstrapping in each of our results tables for the treatment effect of being in an online course. For transparency, online Appendix Table A3 includes our main results table using classroom-level clustered standard errors.

V. Results

In this section, we will review the results from our experiment. First, we present our main results for both the overall course and by individual graded events. Next, we show results by academic ability. Finally, we explore the effects of student relationships with instructors and peers via a end-of-course survey.

A. Main Results

Table 2 presents the main results. In column 1, we show that online instruction reduced a student's final grade by 0.236 standard deviations, or around 1.650 percentage points (out of 100). This result corresponds to about one-half of a $+/-$ grade (e.g., moving from A- to B+). Next, to control for differences in instructor talent, attentiveness, or experience, we add instructor fixed effects to our model. This addition reduces the estimated treatment effect to -0.220 standard deviations, a slight decrease in magnitude.

In the next columns, we add additional fixed effects to control for each level of random assignment. West Point generally randomly assigns students to both the day and hour that they will attend a specific class. First, we add a class day fixed effect that controls for the day of the week and the version of an exam that a student took. This fixed effect is also helpful because most of the instructors are new military faculty who usually teach two sections on one day and two the next day. These fixed effects control for any "learning by doing" for a specific lesson. This specification shows an increased learning gap of 0.223 standard deviations. Next, we control for a class time of day fixed effect that controls for any imbalance across class hour such as West Point's preference to assign NCAA athletes a morning class. This specification decreases the learning gap by 0.005 standard deviations to 0.218 standard deviations. Finally, we add a host of covariates, including whether a student is female, Black, Hispanic, Asian, or an NCAA athlete; attended the prep school; or was previously enlisted in the US Army. This specification, included in column 5, shows a learning gap of 0.215 standard deviations.

These results show gaps in learning and academic achievement between online and in-person instruction and are robust to the addition of instructor fixed effects, class day fixed effects, time of day fixed effects, and covariates. This robustness shows the effectiveness of our experiment and the corresponding balance across class sections and instruction modality. These results are substantial and in line with

TABLE 2—MAIN EFFECTS FOR ONLINE INSTRUCTION

	Final grade (1)	Final grade (2)	Final grade (3)	Final grade (4)	Final grade (5)
Online	−0.236 (0.086)	−0.220 (0.087)	−0.223 (0.086)	−0.218 (0.089)	−0.215 (0.084)
Instructor FEs	No	Yes	Yes	Yes	Yes
Class day FEs	No	No	Yes	Yes	Yes
Time of day FEs	No	No	No	Yes	Yes
Covariates	No	No	No	No	Yes
Observations	551	551	551	551	551
R^2	0.013	0.026	0.026	0.034	0.173
Robust SEs p -values	0.007	0.011	0.010	0.014	0.011
Wild bootstrapped SEs p -values	0.007	0.010	0.009	0.012	0.013

Notes: This table shows the main result for the effect of online learning on college student academic achievement. Covariates include whether a cadet is female, Black, Hispanic, Asian, or an NCAA athlete; attended USMAPS; or was prior enlisted. We also add instructor fixed effects, class day fixed effects, and time of day fixed effects to control for the level of random assignment. Robust standard errors are in parentheses.

other research that examines the effects of technology in the classroom in an experimental setting (e.g., Carter, Greenberg, and Walker 2017; Figlio, Rush, and Yin 2013).

B. *Heterogeneous Treatment Effects by Graded Events and Student Characteristics*

We next examine results for each graded event and student characteristics. These results are helpful because they show the disparate effects of online learning on types of assignments and students. [Figure 1](#) visually shows these heterogeneous treatment effects. We display each point estimate along with its corresponding 95 percent confidence interval and sample size.

Panel A shows results for graded events. The first plot shows the baseline effect from Table 2. Problem sets tested a student's knowledge at the end of each curriculum block. Students were free to work with each other and use resources such as the textbook, but they turned in their own work and reported which classmates they asked for help. We find that students in online classroom scored 0.106 standard deviations lower than their peers in in-person classes. The next estimate reflects daily homework assignments that were worth five points each and are usually viewed as a measure of student engagement and persistence as students earned points solely for completing the assignment. The learning software grades homework automatically, so there is no instructor discretion. We find that online instruction lowered a student's homework grade by 0.211 standard deviations, a result similar to the overall course grade.

Next, we examine the effects for exams. During the course, instructors distributed exams via online learning software regardless of teaching modality. Exams were identical, and all students took exams online in their dorm rooms. There were

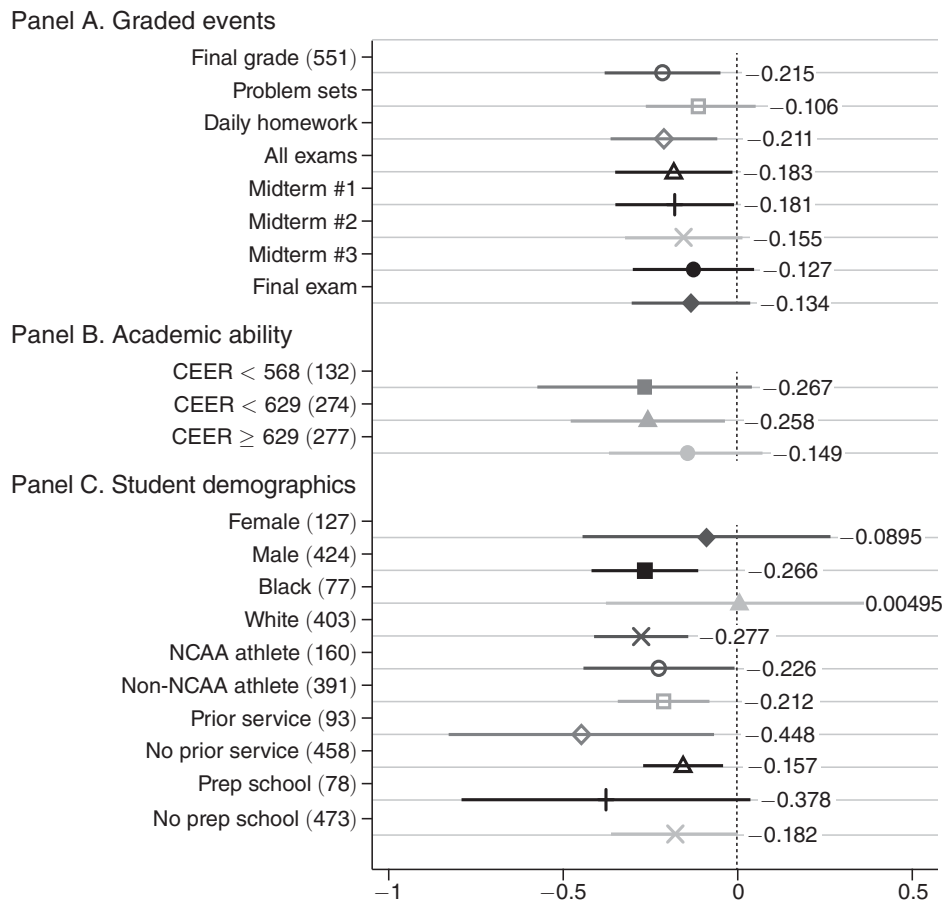


FIGURE 1. HETEROGENOUS TREATMENT EFFECTS OF ONLINE INSTRUCTION

Notes: This figure shows heterogeneous treatment effects of online learning over many dimensions. Each plot corresponds to the indicator for being in an online class (β_1 from equation (1)). Panel A displays the effects for each graded event. The first estimate is identical to column 5 of Table 2. Panel B shows the effect of online learning depending on academic ability given whether a student has a less than lower quartile, less than median, or greater than median CEER score with final grade as outcome. Panel C displays heterogeneous treatment effects for various student demographics with final grade as outcome. All estimates include the same covariates and fixed effects as in column 5 of Table 2. Outcomes are standardized grades.

two versions of the final exam. However, students do not take the final exam with their classmates; instead, West Point randomly assigns exam days and times with no regard to instructor or teaching modality. All students took the final exam online in the instructional software provided by the textbook publisher. For the exam, we added version fixed effects. We find negative and statistically significant effects for exams,⁹ while the gap between online and in-person classes narrows very slightly over time.¹⁰

Panel B presents the results based on CEER scores. Our analysis reveals that students with a CEER score below the twenty-fifth percentile (568) and a median CEER

⁹We standardized the total score a student earned from exams for the “All Exams” results and each midterm individually for the other specifications.

¹⁰We also examined follow-on effects for those cadets who chose to major in economics in online Appendix Table A10. We found small negative effects for switching one’s major to economics and negative, but noisy, reductions in grades for further economics classes. However, we can only identify off very few cadets, and our analysis is underpowered.

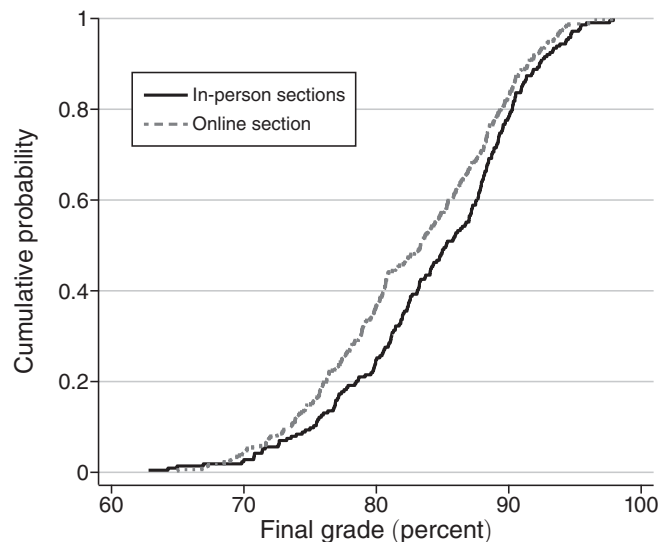


FIGURE 2. CUMULATIVE DISTRIBUTION FUNCTION FOR FINAL COURSE GRADE BY TEACHING MODALITY

Notes: This figure shows cumulative distribution functions for a student's final course grade by teaching modality. The figure plots raw data without controls. The solid line represents grades from an in-person class section, while the dotted line represents online class sections. The figure shows that the bulk of the online mass is among lower final grades and the in-person distribution dominates at all levels.

(629) experience a reduction of 0.267 and 0.258 standard deviations, respectively, when enrolled in an online section. Additionally, students with an above-median CEER score who opt for online learning experience a reduction of 0.149 standard deviations. These findings suggest that online learning has a disproportionate effect on students with lower academic ability.

In Panel C, we examine the results based on student demographics. The most significant effects are observed among male students, White students, prior service individuals, and prep school graduates. These findings indicate that students who traditionally face academic challenges perform worse in an online learning environment. However, we did not observe significant online effects among female students and Black students. Additionally, NCAA athletes performed similarly to the average online student, suggesting that their performance did not suffer more in an online setting.

Analysis by Grade Distribution.—We consider how online education affected students based on their performance in the course. We present these results visually in [Figure 2](#). In this figure, we plot a cumulative distribution function by whether a student was in an online or in-person class section. In the figure, the dotted line represents online class sections and the solid line represents in-person sections. We find that at every point on the distribution (except for a few points on the very bottom), the in-person sections dominate the online sections. In the graph, the mass of the online distribution is located below the mass on the in-person grades. We find that students at the extremes of the grade distribution (i.e., students who earned an A or F) performed similarly, but the gap begins to widen among students at the lower half of the distribution and peaks around the median. The gap then narrows as student performance increases.

TABLE 3—EFFECTS OF ONLINE INSTRUCTION ON POSTCOURSE SURVEY

	Study time (mins.)	Student concentration		Connected to instructor		Instructor cares		Connected to peers	
	OLS (1)	OLS (2)	Ordered probit (3)	OLS (4)	Ordered probit (5)	OLS (6)	Ordered probit (7)	OLS (8)	Ordered probit (9)
Online	2.352 (1.699)	−0.557 (0.100)	−0.692 (0.124)	−0.342 (0.084)	−0.500 (0.121)	−0.126 (0.084)	−0.200 (0.127)	−0.500 (0.119)	−0.534 (0.129)
Observations	402	402	402	402	402	402	402	402	402
R^2	0.104	0.146	0.079	0.179	0.079	0.100	0.046	0.123	0.045
Robust SEs	0.167	0.000	0.000	0.000	0.000	0.134	0.116	0.000	0.000
p -values									
Wild boot- strapped SEs p -values	0.161	0.000	0.000	0.000	0.000	0.121	0.121	0.001	0.000

Notes: This table shows the result for the effect of online learning on how students felt about their course experience from a postcourse survey. “Study time” is how many minutes a student prepared for each class meeting. “Student concentration” is how students rated their in-class concentration on a five-point scale. “Connected to instructor” is how students rated their connection to their instructor on a five-point scale. “Instructor cares” represents how much students felt their instructor cared about them measured on a five-point scale. “Connected to peers” is how students rated their connection to peers on a five-point scale. All estimates include the same covariates and fixed effects as in Column 5 of Table 2. Robust standard errors are in parentheses.

C. Mechanisms: Postcourse Survey

Table 3 shows the effects of online instruction on a host of qualitative outcomes (obtained via a postcourse survey) to address potential mechanisms. First, we ask students for an estimated number of minutes they needed to prepare for each class meeting. Column 1 shows that students in online classes reported studying 2.352 more minutes per day, a result that is statistically insignificant. However, it is interesting that students spent slightly more time studying and yet still received a lower grade.

We next asked students to rank their ability to concentrate in class. Students could respond with a “1” for not being able to concentrate at all up to a “5” if they were perfectly able to concentrate. The average student responded with a “3.515” with a standard deviation of 0.951. We estimate both an OLS regression and an ordered probit to estimate the effect of online instruction on concentration. Columns 2 and 3 show that online instruction lowered student perception of being able to concentrate by 0.557 and 0.692 in the OLS and ordered probit, respectively. These results are roughly three-fourths of a standard deviation decline that is statistically significant.

On our survey, we also explore how online instruction affected the student-instructor relationship. We asked students to rank how connected they felt to their instructor from 1 to 5. On average, students ranked how connected they felt as 3.798 with a standard deviation of 0.831. Columns 4 and 5 show results for these estimates. We find that students rated their connection with instructors lower by 0.342 and 0.500 (OLS and ordered probit, respectively) on the five-point scale. This result translates to around two-thirds of a standard deviation and is statistically significant.

We then asked students whether they felt that their instructor cared about them. Again, students could respond with a 1 if they felt that their instructor did not care about them at all or up to a 5 if they strongly felt their instructor cared. The average student response was a 4.2 with a standard deviation of 0.784. Columns 4 and 5 show these results for both the OLS and ordered probit. We find that being in an online class reduced whether a student felt that their instructor cared about them. We also find that students' perception about their instructors caring about them declines by 0.126 or 0.200 points using the OLS or ordered probit, respectively, a result that is marginally insignificant. This result amounts to a decrease of about a quarter of a standard deviation.

Finally, we asked students to rank their connection to their peers with a similar scale as before. The average student response was a 2.699 with a standard deviation of 1.039. Columns 6 and 7 show the results for the OLS and ordered probit models, respectively. We find that the online environment reduced how connected a student felt toward their peers by 0.500 or 0.534 points, or a decrease of roughly half a standard deviation or 3.5 percentage points.¹¹

Results from the postcourse survey are interesting because the instructor fixed effects are controlling for instructor attributes, teaching styles, and personalities while comparing student perception and connectivity across teaching modalities. These results show that among the costs of online education during the pandemic were student satisfaction, concentration, and many of the intangible benefits a professor provides in an in-person class. Many researchers have shown that instructor quality, characteristics, and intensity of interaction are important for student success, particularly for lower-income and minority students (De Vlieger, Jacob, and Stange 2019; Figlio and Schapiro 2021; Fairlie, Hoffmann, and Oreopoulos 2014; Bettinger and Long 2005; Carrell and West 2010). We find that online instruction during the COVID-19 pandemic may have disrupted these peer and instructor interactions.

VI. Conclusion

This paper makes several findings regarding the efficacy of online learning given our experimental setting of randomized students to online classrooms as a result of the COVID-19 pandemic. We find that online learning reduced final grades by half a $+/-$ grade (e.g., moving from an A $-$ to a B $+$). The online "penalty" is higher for students with lower CEER scores and for students who were White, male, prep school grads, and prior enlisted. These magnitudes are similar to gains from intense individual tutoring (Guryan et al. 2023) or one year of "business as usual" literacy education (Evans and Yuan 2019). We use a postcourse survey to show that online students experienced reduced concentration, felt less connected to their instructors and peers, and claimed that their instructors cared less about them. These results show the limitations of online learning, especially with the little time instructors had to prepare, adjust teaching styles and pedagogy, and port a traditional classroom environment online. From an ethical perspective, we should note that while it is

¹¹ Hardt, Nagler, and Rincke (2022) show that randomized peer tutoring was helpful during the COVID-19 pandemic. This type of intentional peer support network could be helpful to overcome the negative peer connectivity effects in this paper.

academy-wide policy to randomly assign students to classes, we did adjust the final grade of students in online sections according to our findings and prioritize students with lower CEER scores for in-person classes during the 2021 spring semester.

There are some limitations in our West Point setting. First, our student and instructor sample may not be representative of the universe of higher education. West Point students tend to have higher SAT/ACT scores, do not need to worry about employment, and have free room and board. The pivot to online education (and the accompanied recession caused by the pandemic) may have increased college student unemployment and student financial need (Gurantz and Wiegla 2021; Clelan and Kofoed 2017). These extra stressors and insecurities may have caused students to not complete courses or not return for the next semester. Also, West Point requires students to attend classes regardless of instructional modality, and instructors report attendance to commanding officers who ensure compliance. Finally, while there were some outbreaks of COVID-19, West Point's closed-campus environment, close medical attention, and quarantining limited infection, and campus did not close partway during the semester. While these factors may cause a West Point student's experience to differ from the average college student, we argue that our findings may serve as a helpful lower bound of the negative impact of online instruction. Our results may indicate that other college students, particularly those from disadvantaged backgrounds, may have fared worse.

Our results indicate that there are limitations to the effectiveness of online education, especially during a global pandemic. Like many college instructors, those at West Point had little time to prepare lessons and develop pedagogy to more effectively teach online. Our findings are agnostic to whether those teaching styles could mitigate the learning loss that we observe. However, college administrators and higher education policymakers should cautiously consider whether increased online offerings are actually in a student's best interest and the most effective way to deliver a quality college education.

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